ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/341979402

Canines with different pathologies as carriers of diverse lineages of Escherichia coli harboring mcr-1 and clinically relevant ß-lactamases in central Ecuador

Article *in* Journal of Global Antimicrobial Resistance · June 2020

DOI: 10.1016/j.jgar.2020.05.017

CITATIONS
12

READS
226

8 authors, including:



Viviana Albán
University of Washington
10 PUBLICATIONS **52** CITATIONS
SEE PROFILE



Jeannete Zurita
Pontifical Catholic University of Ecuador
104 PUBLICATIONS **1,977** CITATIONS
SEE PROFILE



Rafael Tamayo Trujillo
Universidad UTE
52 PUBLICATIONS **394** CITATIONS
SEE PROFILE



Fernando Xavier Villavicencio
Instituto Nacional de Investigación en Salud Pública
27 PUBLICATIONS **285** CITATIONS
SEE PROFILE

All content following this page was uploaded by William Calero-Cáceres on 18 June 2020.

The user has requested enhancement of the downloaded file.

Journal of Global Antimicrobial Resistance 22 (2020) xxx–xxx

Contents lists available at ScienceDirect

Journal of Global Antimicrobial Resistance

journal homepage: www.elsevier.com/locate/jgar

Letter to the Editor

Canines with different pathologies as carriers of diverse lineages of *Escherichia coli* harbouring *mcr-1* and clinically relevant β-lactamases in central Ecuador

Sir,

It has been considered that companion animals may act as reservoirs for pathogenic bacteria, which is a major concern considering that inadequate management of animal waste or cross-contact may represent potential transmission routes of pathogenic bacteria, including their antimicrobial resistance genes [1]. In this study, we report the presence and diversity of *Escherichia coli* strains harbouring *mcr-1* and clinically relevant β-lactamases isolated from dogs in Ambato, central Ecuador, during 2018.

A total of 117 faecal samples were collected between December 2017 and May 2018 from three veterinary hospitals and three stray dog rescue facilities. Collection of samples was based on the owner's willingness to collaborate in this study. A standardised questionnaire was used to collect animal information. Single colonies with *E. coli*-like morphology isolated from faecal samples were confirmed using biochemical tests. Antimicrobial susceptibility testing was performed using VITEK(2 and the broth microdilution method for colistin according to Clinical and Laboratory Standards Institute (CLSI) guidelines.

Phenotypic extended-spectrum β-lactamase (ESBL), serine carbapenemase and metallo-β-lactamase production was evaluated by double-disc synergy tests. Isolates were screened by PCR and sequencing analysis for the presence of genes conferring resistance to colistin (*mcr-1*), β-lactams (bla_{OXA-48}, bla_{VIM}, bla_{KPC}, bla_{NDM}, bla_{IMI}, bla_{IMP}, bla_{TEM}, $bla_{CTX-M\ groups\ 1,\ 2,\ 8,\ 9\ and\ 25}$, bla_{SHV} and bla_{CMY}), tetracycline (*tetA*) and sulfonamides (*sul1*) using the primers listed in Supplementary Table S1.

The sequence type (ST) of *mcr-1*-positive *E. coli* isolates was identified by multilocus sequence typing (MLST) according to the database of the Center for Genomic Epidemiology. Pulsed-field gel electrophoresis (PFGE) was performed using the restriction enzyme *Xba*I. Comparison of PFGE patterns was performed with GelJ software v.2.0 using the Dice similarity coefficient. Clusters were defined as DNA patterns sharing >85% similarity. Finally, to characterise plasmids related to the *mcr-1* gene, PCR-based replicon typing (PBRT) was performed.

The results revealed that only 3 of the 117 isolates were positive for *mcr-1* (strains UTA-028, UTA-069 and UTA-103), all of which were from veterinary hospital dogs. *Escherichia coli* UTA-028 was isolated from a dog with canine gastroenteritis (a 1-month-old male Shi Tzu) and was resistant to ceftriaxone and ciprofloxacin. In contrast, *E. coli* UTA-069 was isolated from a dog with canine distemper (a 6-month-old female Cocker Spaniel) and showed resistance to ceftriaxone, ceftazidime, ampicillin/sulbactam and ciprofloxacin. Furthermore, *E. coli* UTA-103 was isolated from an animal with no clinical signs of disease (an 8-year-old male mixed-breed) and showed resistance to ceftriaxone, ampicillin/sulbactam, gentamicin and ciprofloxacin (Table 1).

Based on MLST, *E. coli* strains UTA-028, UTA-069 and UTA-103 belonged to ST162, ST1196 and ST744, respectively. The ST162 and ST1196 isolates harboured $bla_{CTX-M-55}$, whilst the ST744 isolate harboured $bla_{CTX-M-65}$. None of the 117 isolates collected were positive for carbapenemase genes. PFGE results revealed two different patterns between strains UTA-069 and UTA-103. Strain UTA-028 could not be typed via PFGE either due to bacterial lysis problems or DNA digestion (Supplementary Fig. S1).

The three strains successfully transferred the *mcr-1* gene to *E. coli* J53, indicating its plasmid origin (Table 1; Supplementary Table S2). Different types of plasmids were found by PBRT. Strains UTA-028 and UTA-069 shared IncFIIS and IncFII plasmids, and all three strains shared an $IncI_2$ replicon-type plasmid. Unfortunately, despite our efforts, we could not determine which plasmid carried the *mcr-1* gene.

In general terms, the occurrence of *mcr-1*-containing *E. coli* isolates was low relative compared with the occurrence in domestic animals in other countries [2]. Some authors suggest

Table 1
Characterisation of *mcr-1*-carrying colistin-resistant *Escherichia coli* isolates (UTA) and transconjugants (CONJ) recovered from dog faeces in central Ecuador.

Strain	ST	Colistin MIC (mg/L)	Plasmid type	Additional antibiotic resistance phenotype	Phenotypic synergies	Additional ARGs
UTA-028	ST162	4	$IncI_2$, IncN, IncFIIS, IncFII	CRO, CIP	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-55}$, *tetA*, *sul1*
UTA-069	ST1196	4	$IncI_2$, IncFIIS, IncFII, $IncX_1$	SAM, CAZ, CRO, CIP	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-55}$, *tetA*, *sul1*
UTA-103	ST744	4	$IncI_2$, IncIγ	SAM, CRO, GEN, CIP	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-65}$, *tetA*, *sul1*
E. coli J53	–	0.5	–	–	–	–
CONJ-028	–	4	–	CRO	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-55}$
CONJ-069	–	4	–	CAZ, CRO	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-55}$
CONJ-103	–	4	–	SAM, CRO	ESBL (+), APB (−), EDTA (−)	bla_{TEM}, $bla_{CTX-M-65}$

ST, sequence type; MIC, minimum inhibitory concentration; ARG, antimicrobial resistance gene; CRO, ceftriaxone; CIP, ciprofloxacin; SAM, ampicillin/sulbactam; CAZ, ceftazidime; GEN, gentamicin; ESBL, extended-spectrum β-lactamase; APB, 3-aminophenylboronic acid; EDTA, ethylene diamine tetra-acetic acid.

http://dx.doi.org/10.1016/j.jgar.2020.05.017
2213-7165/© 2020 The Authors. Published by Elsevier Ltd on behalf of International Society for Chemotherapy of Infection and Cancer. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

that the occurrence of this gene in domestic animals might be related to the consumption of animal-origin food [3]. In the current study, there was a notable difference in terms of clonal diversity in these isolates. Regarding the detected ESBL genes ($bla_{CTX\text{-}M\text{-}55}$ and $bla_{CTX\text{-}M\text{-}65}$), they have been found frequently in *E. coli* of human and animal origin in Ecuador [4,5].

All of the strains mentioned in this study harboured $IncI_2$-type plasmids, which are commonly linked to carriage of the *mcr-1* gene [2]. This observation suggests that acquisition of the *mcr-1* gene by bacteria from domestic animals poses a threat considering their ability for horizontal gene transfer. Colistin is not commonly used in the treatment of domestic animals, therefore the means of colistin resistance acquisition in these strains was from an external source. The presence of multidrug-resistant (MDR) *mcr-1*-carrying *E. coli* in dogs with different clinical diagnoses is worrisome and suggests that pets might represent a silent reservoir of MDR bacteria. These findings highlight the importance of properly managing pet waste and suggest that interactions between people and their pets should be modified in some circumstances.

Funding

This work was supported by the Dirección de Investigación y Desarrollo [UTA HCU-0192-CU-P-2018), Zurita&], Zurita & Zurita Project MIC015 and the Instituto Nacional de Investigación en Salud Pública (INSPI) research funds.

Ethical approval

Not required.

Competing interests

None declared.

Acknowledgments

The authors thank Xiomara Garcés and Vanessa Sailema for their excellent technical assistance.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at https://doi.org/10.1016/j.jgar.2020.05.017.

References

[1] Wieler LH, Walther B, Vincze S, Guenther S, Lübke-Becker A. Infections with multidrug-resistant bacteria—has the post-antibiotic era arrived in companion animals? In: Sing A, editor. Zoonoses—infections affecting humans and animals. Dordrecht, The Netherlands: Springer; 2015. p. 433–52, doi:http://dx.doi.org/10.1007/978-94-017-9457-2_17.

[2] Lei L, Wang Y, Schwarz S, Walsh TR, Ou Y, Wu Y, et al. *mcr-1* in Enterobacteriaceae from companion animals, Beijing, China, 2012-2016. Emerg Infect Dis 2017;23:710–1, doi:http://dx.doi.org/10.3201/eid2304.161732.

[3] Sun J, Zeng X, Li X-P, Liao X-P, Liu Y-H, Lin J. Plasmid-mediated colistin resistance in animals: current status and future directions. Anim Health Res Rev 2018;18:136–52, doi:http://dx.doi.org/10.1017/s1466252317000111.

[4] Loayza-Villa MF, Salinas L, Tijet N, Villavicencio F, Trujillo R, Salas S, et al. Diverse *Escherichia coli* isolates from domestic animal in the same Ecuadorian household all carry the colistin resistance gene *mcr1.1*. J Glob Antimicrob Resist 2019, doi:http://dx.doi.org/10.1016/j.jgar.2019.12.002 [Epub ahead of print].

[5] Ortega-Paredes D, Haro M, Leoro-Garzón P, Barba P, Loaiza K, Mora F, et al. Multidrug-resistant *Escherichia coli* isolated from canine faeces in a public park in Quito, Ecuador. J Glob Antimicrob Resist 2019;18:263–8, doi:http://dx.doi.org/10.1016/j.jgar.2019.04.002.

M. Viviana Albán[1]
Centro de Referencia Nacional de Resistencia a los Antimicrobianos (RAM), Instituto Nacional de Investigación en Salud Pública 'Dr Leopoldo Izquieta Pérez'–INSPI, Quito, Ecuador

Eduardo J. Núñez[1]
UTA-RAM-One Health Group, Centro de Investigaciones Agropecuarias, Facultad de Ciencias Agropecuarias, Universidad Técnica de Ambato (UTA), Ambato, Ecuador

Jeannete Zurita
Unidad de Investigaciones en Biomedicina, Zurita & Zurita Laboratorios, Quito, Ecuador

José E. Villacís
Facultad de Medicina, Pontificia Universidad Católica del Ecuador, Quito, Ecuador

Rafael Tamayo
Centro de Referencia Nacional de Resistencia a los Antimicrobianos (RAM), Instituto Nacional de Investigación en Salud Pública 'Dr Leopoldo Izquieta Pérez'–INSPI, Quito, Ecuador

Gabriela Sevillano
Unidad de Investigaciones en Biomedicina, Zurita & Zurita Laboratorios, Quito, Ecuador

Fernando X. Villavicencio
Centro de Referencia Nacional de Resistencia a los Antimicrobianos (RAM), Instituto Nacional de Investigación en Salud Pública 'Dr Leopoldo Izquieta Pérez'–INSPI, Quito, Ecuador

William Calero-Cáceres[a,b,*]
[a]UTA-RAM-One Health Group, Centro de Investigaciones Agropecuarias, Facultad de Ciencias Agropecuarias, Universidad Técnica de Ambato (UTA), Ambato, Ecuador

[b]Facultad de Ciencia e Ingeniería de los Alimentos y Biotecnología, Universidad Técnica de Ambato UTA, Ambato, Ecuador
[1]These two authors contributed equally to this work.

* Corresponding author at: UTA-RAM-One Health Group, Universidad Técnica de Ambato, Los Chasquis and Río Payamino, Ambato, Ecuador.
E-mail addresses: wcaleroc@hotmail.com,wr.calero@uta.edu.ec (W. Calero-Cáceres).

Received 5 August 2019

Available online 6 June 2020

View publication stats